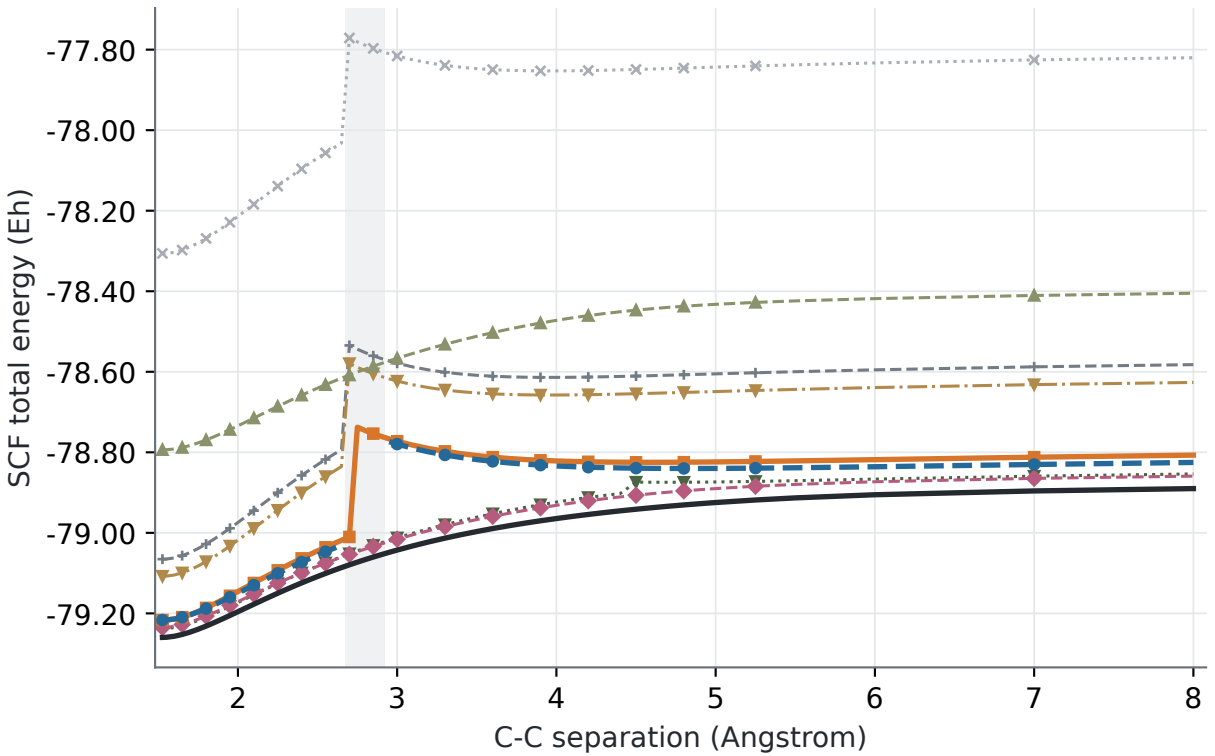
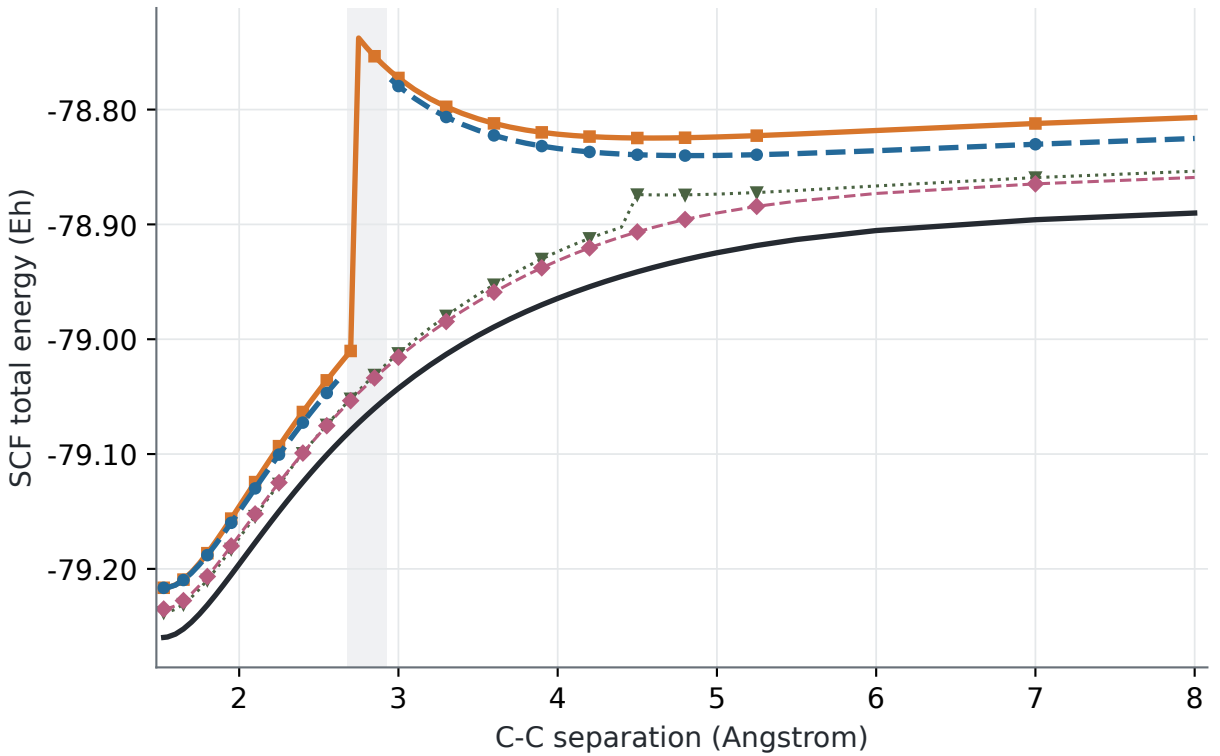
# Ethane C-C stretch: absolute energies

STO-3G (16 AO)
STO-6G (16 AO)
STO-6G + C(d), H(p) (44 AO)
3-21G (30 AO)
6-31G(d,p) (58 AO)
cc-pVDZ (58 AO)
cc-pVTZ (144 AO)
NNAO / MACE (44 AO)
NNAO / free contraction (44 AO)

## All nine curves




## Polarized bases and NNAO




56 rigid geometries; restricted DF-RHF / def2-universal-jkfit; MACE update 3351, fixed exponents and core.
Original Hartree energies; no energy shifts. Gray band: five unconverged free-contraction endpoints (kept as gaps).
Local restricted-SCF solutions; this pressure test is not the physical homolytic dissociation limit.